

Equivalent Expressions

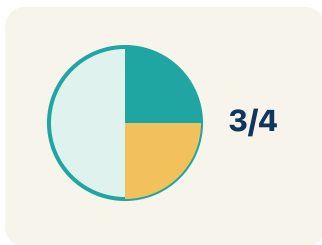
Lesson 6-6

Name: _____ Date: _____ Class: _____

Key Vocabulary

Level 1 support

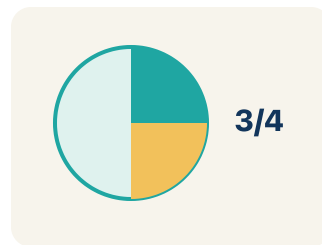
Picture first, then the word, then a plain-language meaning. Say each word out loud.



$2x + 4$ and $2(x + 2)$: when $x = 3$, both = 10; when $x = 7$, both = 18 — always the same

Equivalent

Expressions that always have the same value.



$3x + 2x + 5$ simplifies to $5x + 5$ — fewer terms, same value

Simplify

To write an expression in a shorter, simpler way.

$$3x + 2x = 5x$$

same variable → combine

$3x$ and $7x$ are like terms (both x); $3x$ and $3y$ are NOT (x vs y); $3x$ and $3x^2$ are NOT (x vs x^2)

Like Terms

Terms with the same letter, like $2x$ and $5x$.

$$2 + 3 = 5$$

$4n + 3n = 7n$ — add the coefficients ($4 + 3 = 7$) and keep the variable (n)

Combine

To add or subtract terms with the same letter.

$$4x$$

coefficient = 4

In $5x + 3$, the coefficient of x is 5 — when $x = 2$, the $5x$ part equals 10

Coefficient

The number in front of a letter, like the 3 in $3x$.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can show that two expressions are equivalent by simplifying and combining like terms.

1 Notice

At the music studio, two producers wrote formulas for the cost of renting equipment.

2 Key idea

I can show that two expressions are equivalent by simplifying and combining like terms.

3 Words I'll use

Equivalent

Simplify

Like Terms

Combine

Coefficient



Think About It

- Why did both formulas give the same answer?
- What did Producer B do differently from Producer A?
- Would they still match if $x = 10$?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: Which expression is equivalent to $4x + 3x$?

- A. $7x$
- B. $12x$
- C. $7x^2$
- D. $43x$

Step 1 $4x + 3x = 7x$.

Step 2 Combine the coefficients: $4 + 3 = 7$.



Answer: A. $7x$

 We do — together

Solve this one with your class using the same steps.

Problem: Which expression is equivalent to $2(m + 5)$?

A. $2m + 10$

B. $2m + 5$

C. $m + 10$

D. $7m$

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Which property is shown: $3(x + 7) = 3x + 21$?

A. Distributive

B. Commutative

C. Associative

D. Identity

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Producer A wrote $3x + 2x + 10$ and Producer B wrote $5x + 10$. They look different but both give 30 when $x = 4$. Why are they equivalent?

Level 1 support

Start here: $3x + 2x$ combines to $5x$, so the two forms are equivalent.

 **Need a hint?**

Sentence starters (optional)

These expressions are equivalent because $3x + 2x$ ____.

Estas expresiones son equivalentes porque $3x + 2x$ ____.

I checked by substituting $x =$ ____ and getting ____ both ways.

Lo comprobé sustituyendo $x =$ ____ y obteniendo ____ en ambas.

Word bank: Equivalent Simplify Like Terms Combine

Listen for: Students explain $3x + 2x = 5x$ (combining like terms), so both equal $5x + 10$, and verify by substituting a value like $x = 4$ to get 30 both ways.

Level 2

Is testing only $x = 4$ enough to be sure two expressions are equivalent? Justify why or why not.

- Testing one value is ____ enough because ____.
- To be sure they are equivalent, I need ____.

EXPLORE

How can you prove $2(n + 4)$ and $2n + 8$ are equivalent? What two strategies could you use?

Level 1 support

Start here: Distribute or substitute values to show the forms always match.



Need a hint?

Sentence starters (optional)

They are equivalent because distributing $2(n + 4)$ gives ____.

Son equivalentes porque distribuir $2(n + 4)$ da ____.

I can also check by substituting $n =$ ____ and getting ____.

También puedo comprobar sustituyendo $n =$ ____ y obteniendo ____.

Word bank: **Equivalent** **Simplify** **Combine** **Like Terms**

Listen for: A strong answer uses the distributive property ($2(n + 4) = 2n + 8$) and/or substitutes a value into both to confirm equal results.

Level 2

Are $2x + 3$ and $5x$ equivalent? Find a value of x that proves your answer and explain why one test value showing equality is not enough.

- They are ____ equivalent because at $x =$ ____ they give ____ and ____.
- One matching value is not proof because ____.

CONNECT

Client A pays $4h + 2h + 30$ and Client B pays $6h + 30$. Is the pricing fair (equivalent)? How can you prove it?

Level 1 support

Start here: Combine $4h + 2h$ to $6h$, showing both clients pay the same.

 **Need a hint?**

Sentence starters (optional)

The expressions ____ equivalent because $4h + 2h =$ ____.

Las expresiones ____ equivalentes porque $4h + 2h =$ ____.

I can prove it by substituting $h =$ ____: both equal ____.

Puedo probarlo sustituyendo $h =$ ____: ambas dan ____.

Word bank: Equivalent Simplify Combine Like Terms

Listen for: Students simplify $4h + 2h + 30$ to $6h + 30$, conclude the pricing is fair/equivalent, and verify by substituting a value of h .

Level 2

A third client pays $5h + 30$. Critique whether that is the same deal, and explain for which client it is cheaper as hours grow.

- $5h + 30$ is ____ the same because ____.
- As hours grow, the ____ client pays less because the coefficient of h is ____.

Write About the Math

The Writing Revolution

I can explain my work using the words equivalent, simplify, like terms, and combine.

1. Kernel Sentence subject + verb

Model: Equivalent is expressions that always have the same value.

Equivalente es expresiones que siempre tienen el mismo valor.

Write a kernel sentence about equivalent. Use a subject and a verb.

Escribe una oración base sobre equivalente. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Equivalent matters in math

Equivalente importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Equivalent matters in math because ____.

Equivalente importa en matemáticas porque ____.

but
pero

Equivalent matters in math, but ____.

Equivalente importa en matemáticas, pero ____.

so
entonces

Equivalent matters in math, so ____.

Equivalente importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about equivalent.
Di un hecho verdadero sobre equivalent.

Equivalent ____.

Question
Pregunta

Ask a question about equivalent.
Haz una pregunta sobre equivalent.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about equivalent.
Muestra entusiasmo sobre equivalent.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with equivalent.
Dile a un compañero qué hacer con equivalent.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

These expressions are equivalent because $3x + 2x$ ____.

Estas expresiones son equivalentes porque $3x + 2x$ ____.

I checked by substituting $x =$ ____ and getting ____ both ways.

Lo comprobé sustituyendo $x =$ ____ y obteniendo ____ en ambas.

They are equivalent because distributing $2(n + 4)$ gives ____.

Son equivalentes porque distribuir $2(n + 4)$ da ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Are $4x + 2$ and $2(2x + 1)$ equivalent?

- A. Yes, they are equal for every value of x
- B. No, never
- C. Only when $x = 0$
- D. Only when $x = 1$

Show your work:

2. Which expression is NOT equivalent to $5x + 10$?

- A. $5x + 5$
- B. $5(x + 2)$
- C. $10 + 5x$
- D. $x \cdot 5 + 10$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A student says $2x + 3$ and $5x$ are equivalent because both equal 5 when $x = 1$. Is the student correct? Explain why testing one value is not enough to prove equivalence, and find a value of x where they give different results.

Sentence starter: The student is ____ because _____. When $x = 1$: $2(1) + 3 = \underline{\hspace{1cm}}$ and $5(1) = \underline{\hspace{1cm}}$. But when $x = \underline{\hspace{1cm}}$: $2(\underline{\hspace{1cm}}) + 3 = \underline{\hspace{1cm}}$ and $5(\underline{\hspace{1cm}}) = \underline{\hspace{1cm}}$. To prove equivalence, expressions must be equal for ____ values of x .

Show your work:

Reflect — Exit Ticket

Which expression is equivalent to $6n + 4 + 3n - 1$?

- A. $9n + 3$
- B. $9n + 5$
- C. $63n$
- D. $18n + 4$

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $2m + 10 - 2(m + 5) = 2m + 10$ using the distributive property.
2. **You Try (notes):** A. Distributive — *Multiplying 3 by each term inside the parentheses is the distributive property.*
3. **Try It 1:** A. Yes, they are equal for every value of x — $2(2x + 1) = 4x + 2$, so they are equivalent for all x .
4. **Try It 2:** A. $5x + 5 - 5x + 5$ is different; the others all equal $5x + 10$.
5. **Exit Ticket:** A. $9n + 3 - 6n + 3n = 9n$ and $4 - 1 = 3$, so the simplified expression is $9n + 3$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Equivalent is expressions that always have the same value.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).